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Key Facts You Should Know about Water

CGTN

- 1 Today, about 2.1 billion people don't have access to safe water, or about 1 in 4 people, according to UNICEF (United Nations International Children's Emergency Fund). Meanwhile, 159 million still drink untreated water — a serious health risk — from surface water sources, such as streams or lakes.
- 2 The average American uses 80–100 gallons of water a day, 10 times more than the average person in a rural community in sub-Saharan (撒哈拉沙漠以南的) Africa.
- 3 An average one-minute shower uses more water (at least 5 gallons) than most people in sub-Saharan Africa use in an entire day for basic drinking and hygiene purposes (2–5 gallons).
- 4 According to WHO (World Health Organization), over 5,000 people die each year, or 14 people daily, in Europe from diarrheal (腹泻的) disease due to inadequate water and sanitation (卫生).



- 5 Fourteen of the world's 20 megacities (大城市) are now experiencing water scarcity or drought conditions, including China's Beijing and Shanghai. As many as four billion people already live in regions that experience severe water stress for at least one month of the year, according to a 2016 study. Nearly half of those people live in India and China.
- 6 More than 800 million people need to travel and queue for at least 30 minutes to access safe supplies.
- 7 Gender is also a key factor because women take the primary burden of collecting water. The time-consuming task of fetching the UN-recommended 50 liters of water per day for a family of four takes the equivalent of two-and-a-half months each year.

8 The proportion of the world's population with access to clean water near their home has risen from 81 to 89 percent since 2000. China, which has made the greatest progress, has seen an extra 334 million people get access to water between 2000 and 2015.

Group work: Work in groups and discuss the following questions: Which fact impresses you the most? What measures can be taken to make more people have access to safe water?



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Pair Work: Watch the video clip twice and complete the following sentences according to what you have heard. Compare notes with your partner. The words below may be new to you.

sanitation /,sæni'teɪʃən/ *n.* 卫生

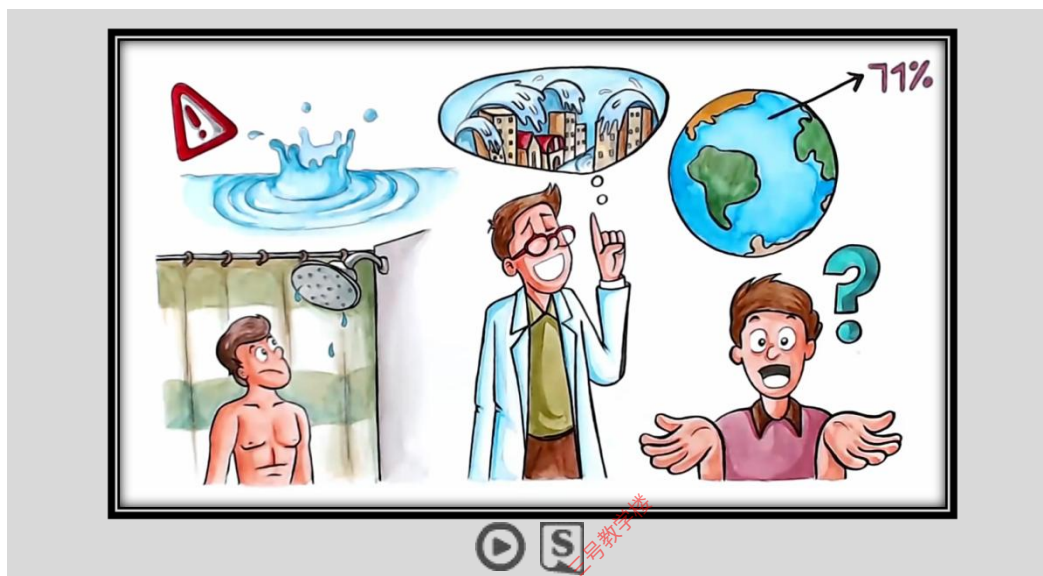
component /kəm'pəʊnənt/ *n.* 组成部分

livestock /'laɪvstɒk/ *n.* 家畜

1 The speaker is talking about the coming global water crisis/water problem .

2 Water is the key component in our daily life, not only for human use , but for energy, industry, agriculture , and livestock.

3 If you look at the Earth , you will notice that over 75% of our blue planet is water. But the problem is only 3% of that water is fresh water .



Watch the video clip and discuss the following questions with your classmates.

- 1 According to the video clip, what percentage of the Earth surface is covered by water?
- 2 What percentage of the water all over the world is fresh water in liquid form?
- 3 On average, how many people do not have access to safe water?
- 4 What human activities contribute to water pollution?

Kennedy's moon-shot speech

On May 25, 1961, while speaking to Congress, US President John F. Kennedy challenged Congress and the American people to put a man on the moon, and return him safely to Earth by the end of the decade. The speech is later known as “the moon-shot speech”.

美国总统约翰·F·肯尼迪于1961年5月25日在国会发表演说时，向美国国会和人民提出挑战，要求在十年内完成登月计划并让宇航员安全返回地球。这篇演说后来被称为“登月演说”。



Hydroelectric dam

Hydroelectric power is produced as water passes through a dam, and into a river below. The more water that passes through a dam, the more energy is produced. The Chinese Three Gorges Dam is the world's largest power station in terms of installed capacity (22,500 MW).

水力发电的原理是利用水流流经大坝、流入下游河道时所产生的势能，推动发电机，进而转化为电力。流经大坝的水量越多，转化而成的电力越多。我国三峡大坝是当今世界最大的水力发电工程。



TEXT

Water shortage is likely to be one of the greatest problems for the first half of this century. Yet solving the water problem is not going to happen overnight. It will take time. Nonetheless, there are some things we can start on straight away. As Michael Webber argues, we need aggressive conservation to buy us time before scientists come up with affordable solutions.

What's the World to Do about Water?

Michael E. Webber



 **T** 1 In 2007, when my daughter was seven years old, we would brush our teeth together every night as part of our daily ritual. To **conserve** water, we would turn off the faucet after wetting our brushes and turn it back on only to rinse. One night, I didn't turn off the water fast enough **to her liking**. She turned off the faucet, made an angry face at me, and growled, "Turn off the water, Daddy. The scientists need time."

 **T** 2 That statement still **rings in my ears** today, not only because of her precociousness but also because she was exactly right. We need to conserve our **resources** to buy ourselves time so scientists can find new solutions to our problems. And this is especially true for water.

 **T** 3 Water is **on track** to be the most important and most contentious resource of the 21st century. It could **replace** oil as the strategic resource that triggers geopolitical **conflicts**. But with the right **solutions**, it could also be the one that brings us all together.

- 4 In 2003, the late Nobel Laureate Richard Smalley gave a lecture at Rice University's Energy & Nanotechnology Conference **highlighting** humanity's top 10 problems for the next 50 years. His list, in descending order of importance, was: energy, water, food, **environment**, poverty, terrorism and war, disease, education, democracy, and population.
- 5 The reason energy and water sit at the top, ahead of food and poverty, is that addressing them makes **subsequent** problems easier to deal with. Developing **abundant sources** of clean, **reliable**, affordable energy enables an abundance of clean water. An abundance of clean water **enables** food production and protects the environment. **And so forth.**
- 6 In many ways, the 1900s was the century of energy conflict. Now, the dawn of a new energy era is just **around the corner** — with the prices of solar **power going down**, and **distributed** generation and energy **efficiency on the point of** taking off — **and we can foresee** a time this century when water replaces energy as the next great challenge for humanity. Getting water right could clear the path to a fully liberated, **healthy**, and peaceful **civilization.**

 **T** 7 But water is **complicated**. First, there's no **alternative**. There are alternatives to coal and petroleum, but water cannot be replaced. Second, water is intertwined with every other sector of society. Energy production requires water for cooling power plants and fracturing shale. Agriculture needs water for irrigation. Industry and cities use trillions of gallons of water for all sorts of purposes. Third, water demands are growing at the same time supplies are fluctuating. Climate change is expected to intensify droughts and floods while shifting where water will be and when. That means humanity will **be confronted with** decisions about whether to move people to water or water to people.

 **T** 8 Thankfully, there are solutions. Some are large, incredibly expensive, and energy-intensive, such as building cross-continent water-transfer aqueducts, new hydroelectric dams, and massive desalination plants. Some are small, such as using micro water harvesters that condense water vapor out of the air. All of them take a while.

  9 In April 1961, President John F. Kennedy said, “If we could ever competitively — at a cheap **rate** — get freshwater from saltwater, it would be **in the** long-range **interest of** humanity and would really **dwarf** any other scientific accomplishments.” That was one month before his famous moon-shot speech. Before the decade was over, we sent man to the moon and back. But we still haven’t found a solution as to how to get freshwater from saltwater without a lot of money and energy. That’s mainly because our great **innovation** apparatus has not been dedicated to water. Research **budgets** have prioritized energy, defense, and health while **mostly** ignoring water, even though fixing water would help our energy, defense, and health problems. If we spent as much money looking for water on Earth as we do searching for it on Mars, the **outcomes** might be very different.

🔊 T 10 In the end, we can solve the water problem. But we need aggressive conservation that will buy us time while our inventors get to work.



1. Text Organization

In this text, the author does not tell us a story. Nor does he describe an event. Instead, he tells us what he thinks of something, and then explains why he thinks so. Let us go over the paragraphs and see how he does that.

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1.1 First, find out what the following paragraphs are about.

Paras. 1–2 We need to conserve water aggressively so that scientists can have time to find solutions

Paras. 3–7 Water is going to be the most important and most contentious resource of the 21st century

Paras. 8–9 There are solutions, but they take time

Para. 10 Aggressive conservation is needed to buy us time before scientists find solutions

1.2 Answer the following questions:

 1 What is the central idea of the text?

Water has become one of the greatest and complicated problems the world is faced with today, which needs us to take action right now to conserve resources so as to give scientists enough time to find solutions.

 2 How does the author raise the water problem at the beginning of the article? Why does he do it in this way?

To introduce the water problem, the author tells us about what happened one night when he and his seven-year-old daughter were brushing teeth together. In so doing the author tries to introduce the reader to a serious topic in a casual manner. Besides, it also reminds us of the fact that everyone can play a role, even in our daily life, in saving water.

3 How does the author try to prove that water is on the top of the list of the world problems today?

He cites the late Nobel Laureate Richard Smalley's list of humanity's top 10 problems for the next 50 years.

4 What is the purpose of the author in citing President John F. Kennedy's words?

He wants to show the reader how difficult it is for us to find a solution to the water problem.

5 How does the author bring his writing to an end?

He repeats what he says at the beginning of the article: The water problem can be solved but "The scientists need time."

2. Comprehension Check

2.1 Focusing on the main ideas

Complete the following passage with key information from the text.

In his 2003, the late Nobel Laureate Richard Smalley highlighted the top 10 problems for the next 50 years. On his list, water comes second. However, as time goes by, the water problem will be most likely to become the greatest challenge for humanity today.

This is mainly because water problem is complicated. First, unlike coal and oil, water cannot be replaced. Second, water is intertwined with every other sector of society. Third, while water demands are growing, climate change is intensifying droughts and floods and water supplies are fluctuating.

True, there are some possible solutions, but they may be incredibly expensive. Anyway, it will take time for scientists to find cost-effective solutions to the problem. Before that, everyone should do his or her bit to conserve water.



2.2 Digging into detail

Answer the following questions to better understand the text. You may want to write down some key words in the space provided before you do so.

🔑 *(Para. 3)* **1** What does the author mean by saying that water will most probably become the most contentious resource of the 21st century?

He means that water is most likely to become a strategic resource like oil that could trigger geopolitical conflict.

🔑 *(Paras. 4-5)* **2** On Prof. Smalley's list of the problems for the first half of this century, energy and water sit at the top. Why is it so?

Because solving the energy and water problems will make the other problems easier to deal with. Abundant energy and water supplies will increase food production and improve the environment. They will also reduce poverty, disease, and so forth.

🔑 (Para. 5) 3 What does “affordable energy” most probably mean?
It most probably means energy produced at a low cost.

🔑 (Para. 7) 4 What does the author mean when he says “water is intertwined with every other sector of society”?

He means that every other sector of society needs water. People need water in their daily life. Energy production requires water for cooling power plants. Industry needs water for all sorts of purposes. Agriculture needs water for irrigation.

🔑 (Para. 9) 5 Did President Kennedy realize the importance of finding a cheap way to get freshwater from saltwater? How do you know?

Yes, he did. Because he said getting freshwater from saltwater at a cheap price would dwarf all other scientific accomplishments.

2.3 Understanding difficult sentences

Choose the answer that correctly interprets the sentence from the text.

 1 One night, I didn't turn off the water fast enough to her liking. (*Para. 1*)

- ☒ A) One night, she wasn't pleased with the way I turned off the water, because she thought I didn't do it fast enough.
- ☐ B) One night, I didn't turn off the water as fast as she did; it seemed

 2 We need to conserve our resources to buy ourselves time so scientists can find new solutions to our problems. (*Para. 2*)

- ☐ A) We should conserve our resources. If we do so the money we save will be used to find new solutions to our problems.
- ☒ B) We should conserve our resources. If we do so the resources will probably last long enough for scientists to find new solutions

to our problems

🔑 3 His list, in descending order of importance, was: energy, water, food, environment, poverty, terrorism and war, disease, education, democracy, and population. (*Para. 4*)

- ☒ A) His list contains ten items (项): energy, water, food, environment, poverty, terrorism and war, disease, education, democracy, and population. They are arranged in such a way that each item is more important than the one that follows it.
- ☐ B) The ten items on his list — energy, water, food, environment, poverty, terrorism and war, disease, education, democracy, and population — are gradually losing importance now.

 **4** The reason energy and water sit at the top, ahead of food and poverty, is that addressing them makes subsequent problems easier to deal with. (*Para. 5*)

- ☒ A) Energy and water are the top two problems on the list. The reason is that if we start to solve these problems, it will be easier to deal with the other problems that follow them on the list, such as food and poverty.
- ☐ B) Energy and water are listed as the two problems more important than food and poverty. This is because once we start working on them, it will be easier to address food shortage and poverty later on.

🔑 5 ... even though fixing water would help our energy, defense, and health problems. (*Para. 9*)

- ☒ A) ... even though solving the water problem would help solve energy, defense and health problems.
- ☐ B) ... even though using water in an economical (节约的) way would help solve energy, defense and health problems.

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The writer tries to explicate the topic of water conservation in this expository essay. He develops his argument to inform and persuade the reader through the complete structure of an introduction, body and conclusion.

In the introductory paragraphs, he adopts a light story-telling style to make his serious point clear and accessible. Here the conflict between him and his daughter draws the reader in, and then quickly the thesis factors in to the climax, where the reader hears the resounding words of his daughter: “Turn off the water, Daddy. The scientists need time.” The same message is rephrased in the second paragraph: “We need to conserve our resources to buy ourselves time so scientists can find new solutions to our problems.” Such a repetition of the same message helps to hit the nail home.

In the main body of the text, he employs a variety of techniques to develop his argument. Skillfully, he helps readers to think logically about the water problem and to easily grasp its importance and the reasons why water conservation is imperative.

The first technique is *dividing and classifying*. He cites a Nobel Laureate's work to divide humanity's top issues for the next 50 years into ten problems in order of importance (Para. 4). Then he classifies the problems by putting water in the top tier (Para. 5) and thus explains the gravity of the water problem with clarity. Also, he divides solutions to the water problem in terms of the scale of the projects (Para. 8).

The second technique is *explaining cause-and-effect relationships*. It is necessary to help readers see and understand these relationships involved in the water problem: What have caused the water problem (Para. 7 & Para. 9)? What benefits can solving the water problem bring (Para. 3, Para. 5 & Para. 6)?

The third technique is *comparing and contrasting*. In discussing solutions to the freshwater problem, the writer contrasts the two types of solutions for differences, and also compares them to bring up the argument that both take time (Para. 8). Similarly, he compares and contrasts the moon shot mission and getting freshwater to make a case for more research funding for the latter (Para. 9).

How can we help relieve the problem of water shortage? And what can we do to achieve a balance between economic growth and resource conservation? You may discuss these issues from different perspectives (the government, factories, individuals ...).



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conserve	conservation	resource	replace	conflict
highlight	environment	subsequent	abundant	source
reliable	enable	distribute	efficiency	foresee
civilization	complicated	alternative	intensify	rate
dwarf	innovation	budget	mostly	outcome
to one's liking	ring in one's ears	on track	and so forth	around the corner
go down	on the point of	be confronted with		in the interest of





1. Fill in the blanks with words or phrases from the box below. Change the form where necessary.

replace	conflict	highlight	environment	subsequent
abundant	rate	source	reliable	distribute
foresee	civilization	intensify	complicated	budget
innovation	mostly	outcome	on the point of	
be confronted with		in the interest of		ring in one's ears

- 1 It will be a close game (势均力敌的比赛). The outcome is hard to predict.
- 2 Travelers on a tight budget would choose to go there by train instead of by plane.
- 3 Tom was on the point of leaving the garden when he heard someone cry: "Isn't it Tom? I can hardly believe my eyes."
- 4 Innovation is what we need when we design new products.



replace	conflict	highlight	environment	subsequent
abundant	rate	source	reliable	distribute
foresee	civilization	intensify	complicated	budget
innovation	mostly	outcome	on the point of	
be confronted with	in the interest of		ring in one's ears	

5 The problem turned out to be far more complicated than he had foreseen .

6 The works in this room are mostly by local artists. They embody the artists' vision of a peaceful and environment-friendly civilization .

7 Water, though abundant in Brazil, is extremely unevenly (不均匀地) distributed among its regions (地区).

8 Even today, my first English teacher's encouraging words still ring in my ears .

9 Henry has many friends working in the government. His sources of information are usually reliable .



replace	conflict	highlight	environment	subsequent
abundant	rate	source	reliable	distribute
foresee	civilization	intensify	complicated	budget
innovation	mostly	outcome	on the point of	
be confronted with		in the interest of		ring in one's ears

- 10** These figures show that the death rate of road accidents gets higher as the speed of the car increases.
- 11** With water demand growing, the conflict between the two neighboring villages over water has intensified over the years.
- 12** Years ago the professor said that there was no doubt that the people in this city would be confronted with a water shortage, if not a water crisis. Subsequent events have proved his predictions to be exactly right.
- 13** The TV program highlights the race problem in American society.



replace	conflict	highlight	environment	subsequent
abundant	rate	source	reliable	distribute
foresee	civilization	intensify	complicated	budget
innovation	mostly	outcome	on the point of	
be confronted with		in the interest of		ring in one's ears

14 The next few paragraphs will discuss how many jobs have been replaced by technology and other related data to the topic.

15 We need to find ways of producing energy without destroying the environment.



2. Replace the underlined part in each of the following sentences with a word or phrase from the box below. Write your answer in the brackets. Make changes or additions where necessary.

enable dwarf on track and so forth around the corner

- 1 You see the lights in the students' reading room? They never go out at night these days because the final exams are drawing near. (**around the corner**)
- 2 At that time the working class was too small to make a successful revolution there. (**enable**)
- 3 The high-rise building in the back makes our old house seem small. (**dwarfs our old house**)



enable dwarf on track and so forth ~~around the corner~~

- 4 The figures show that our economy is healthy and we are likely to achieve sustainable growth. (**on track**)
- 5 The two girls spent the whole night planning the summer events, get-togethers, dinner parties, and many other things of the kind. (**and so forth**)



3. Complete the following sentences after the model with the word(s) or phrase in brackets. Make changes or additions where necessary.

Model:

That statement still rings in my ears today. (ring in one's ears)

- 1 We believe that building a powerful navy is in the interest of
our national defense. (in the interest of)
- 2 He thought the coffee at breakfast was too sweet. It wasn't
to his liking. (to one's liking)
- 3 The new system is said to enable us to use much less
oil than before, in other words, to increase energy
efficiency considerably. (increase, efficiency)



- 4 Speaking of the real estate (房地产) market, when people are losing their jobs, it is almost certain that the house prices will go down . (**go down**)
- 5 A regional conference on the conservation of fishery (渔业) resources will be held in the international convention (会议) next week. (**conservation**)

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1. first, at first

first: *a.* & *ad.* happening or coming before all other similar things or people 第一, 首先

at first: (used when you are talking about what happens in the early stage of an event, but it is different from what happens later) in the beginning 起先, 最初





e.g.

- 1) It's my **first** long trip without my parents, and I am actually quite excited about it. (*Reading 2, Unit 2*)
- 2) But water is complicated. **First**, there is no alternative. (*Para. 7*)
- 3) **At first** in New York, his uncle in the Bronx offered him shelter while he worked as a restaurant busboy. (*Reading 1, Unit 1*)
- 4) **At first**, two Western governments competed fiercely for the supervision of the project, but neither of them got the upper hand. In the end, they had to let the Chinese do it themselves. (*Reading 1, Unit 4*)



Fill in the following blanks with “first” or “at first” as you see fit.

- 1 Labor Day in the US falls on the first Monday in September.
- 2 The speaker seemed a little nervous at first , but after a short while became relaxed (放松).
- 3 You want to learn making Chinese dumplings (饺子)? First of all, let me show you how to make the filling.
- 4 It was sunny but cool when we started. So the journey was quite pleasant at first . Then all of a sudden there came a thunderstorm.



2. source; resource

source: the place where a stream or river starts 源头
a person, place, thing or activity from which one
gets sth. 来源

resource: (*usu. pl.*) sth. that can be used for support or help 财力,
物力
sth. that exists in a country and can be used to increase
its wealth 资源

e.g.

- 1) We need to conserve our **resources** to buy ourselves time so scientists can find new solutions to our problems. (*Para. 2*)
- 2) Developing abundant **sources** of clean, reliable, affordable energy enables an abundance of clean water. (*Para. 5*)



Fill in the following blanks with “resource(s)” or “source(s)” as you see fit.

1 Canada is a country rich in natural resources .

2 The travelers walked upstream (往上游) until they reached the source of the river.

3 So far there has been no response from official sources .

4 With such limited resources , the manager has been doing remarkably well to keep the organization going.





1. Rewrite the following sentences, using a sentence pattern from one of the boxes below.

the reason (that/why) ...

e.g. *The reason* energy and water sit at the top, ahead of food and poverty, is that addressing them makes subsequent problems easier to deal with.
(Para. 5)

the fact that ...

e.g. Less well-known, however, was *the fact that* the “Aryan” girl it claimed to show was actually Jewish.
(Text, Unit 3)

🔑 1 He has fully recovered from his bad accident. This shows that medicine can work wonders.

The fact that he has fully recovered from his bad accident shows that medicine can work wonders.





🔑 2 Why are businesses moving their plants to developing countries?

It's because labor costs in their home countries are too high.

The reason (that/why) businesses are moving their plants to developing countries is that labor costs in their home countries are too high.

🔑 3 Our rivers are running dry or being polluted. If we go on ignoring this, the consequences (后果) will not be hard to predict.

If we go on ignoring the fact that our rivers are running dry or being polluted, the consequences will not be hard to predict.

🔑 4 Last year's crop wasn't good. That's why food prices are rising.

The reason (that/why) food prices are rising is that last year's crop wasn't good.



2. Rewrite the following sentences, using the sentence pattern from the box below.

whether ... or ...

e.g. That means humanity will be confronted with decisions about

whether to move people to water *or* water to people. (Para. 7)

🔑 1 Did the chicken come before the egg, or the egg before the chicken? This is a question of how life began in general.

Whether the chicken came before the egg, or the egg before the chicken, is a question of how life began in general.



 2 I saw her on the point of tears when she got the result of the exam. Should we go to comfort her? Or should we just leave her alone? I don't know.

I saw her on the point of tears when she got the result of the exam. I don't know whether we should go to comfort her or just leave her alone.

 3 Does she still want to go to the lake? Or has she changed her mind? It's not clear.

It's not clear whether she still wants to go to the lake or has changed her mind.

 4 Should young married couples buy or just rent an apartment? Many are wondering.

Many young married couples are wondering whether they should buy or just rent an apartment.



Pair work: Work with your partner to complete the following dialogues, using the words and phrases in brackets. Make changes or additions where necessary.

Model:

A: What do you think will be the greatest challenge of our time?

B: I think water is on track to be the greatest challenge of our time. It will replace oil as the No. 1 problem of the 21st century. **(replace, on track)**





1 A: Why do you think oil will no longer be the No. 1 problem of our time?

B: The dawn of a new energy era is just around the corner. Oil prices are going down because solar power is becoming cheaper. Besides, improvements in energy efficiency, while helping to create a(n) clean environment, are reducing waste and the demand for oil. **(go down, environment, around the corner, efficiency)**

2 A: Why isn't it easy to solve the water problem?

B: Because water is far more complicated than energy. No other resource can replace water, so it is impossible to find an alternative to it. Besides, water demand is rising, but water supply is fluctuating as a result of climate change. **(resource, complicated, alternative, replace)**



3 A: Why is it important for us to conserve water?

B: Water is going to be the most contentious resource of the 21st century. It will trigger geopolitical conflicts. That's why water conservation is not only in the interest of our economy but also a task of strategic importance. **(conservation, in the interest of, conflict, resource)**

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In the following articles, Reading 1 continues our look at the problem of water scarcity, outlining its effects on health, hunger, education and poverty. Reading 2 describes China's water situation and praises the country's sustainable, long-term solution to its water problem.

Reading 1

Global Water Scarcity

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🔊 T 1 For many people, water has never been a big story in their lives. This is because they live in communities that have good water supply systems. They turn on the tap and clean water flows, every day of the year. This makes it very difficult for people to appreciate how precious water is.

🔊 T 2 Water is life. Plants, animals and humans all depend on this invaluable natural resource for life. Besides this, water is used in cleaning and sanitation, manufacturing, construction and farming. Almost every human activity you can think of involves some use of water.

Water covers more than 70% of the Earth's surface, so how can there ever be scarcity?

🔊 T 3 (1) Less than 3% of water on Earth is fresh water, and the bulk of this is trapped in snowfields and glaciers and not easily accessible. The rest form the seas and oceans and cannot be used in the same way as fresh water. Only a tiny fraction (0.014%) is surface water in the form of rivers, lakes and swamps.

🔊 T 4 Naturally, the 3% should be enough for all humans and animals on Earth, (2) but unfortunately, many factors have caused a major upset in the flow and use of fresh water and have caused massive crises in many regions of the Earth.

But why should you care?

🔊 T 5 We should care because a lot of the factors such as population expansion, urbanization, pollution and climate change that cause water scarcity are broadening and becoming more complex and uncontrollable. (3) This means if we do nothing in terms of preserving and using water wisely, it is only a matter of time before all regions begin to experience water crises and all the repercussions that come with it.

What are the effects of water shortages?

🔊 T 6 The effects of water scarcity can be grouped into these 4 broad areas —Health, Hunger, Education and Poverty.

Health

- 🔊 T 7 In many developing countries, people are forced to drink low quality water from flowing streams, many of which are contaminated. There are many water-borne diseases that people die of.
- 🔊 T 8 Lack of water or quality water causes huge sanitation issues. Clinics, restaurants, public places of convenience and many other places are forced to use very little water for cleaning. This risks the health of the staff and people who use the facilities.

Hunger

- 🔊 T 9 It takes a lot of water to grow food and care for animals. Experts say that globally we use 70% of our water sources for agriculture and irrigation, and only 10% on domestic uses.
- 🔊 T 10 Less water means crops that need water to grow have lower yields. It means farm animals will die and others will not do well without water. The result is constant hunger and thirst and a low quality of life.

Education

- 🔊 T 11 At first sight it is a bit hard to see how water and education are related. Yet in many parts of the world children have to be up at dawn to collect water for the family. They have to walk for several miles to get water. The children get tired and some have to miss school as a result.

Poverty

- 🔊 T 12 Access to quality water is key to economic prosperity and better living standards. Manufacturing, farming, and mining all need a lot of water to thrive. Lack of water means no economic activities will happen and the people will be in constant poverty.

What is your role in water preservation?

- 🔊 T 13 Sometimes the magnitude of a problem can make one feel that there is nothing that can be done. But there is a lot you can do for water. There is a high chance that people reading this do not live in water deprived areas, and may think it is not

their problem. Here is what you can do to help

Awareness

- 🔊 T 14 Learn about the water crisis, just like you are doing. If you understand a problem, you are in a better position to have a solution. Talk about it with family and friends. Look out for news and facts on water shortages and crisis areas.

Take part

- 🔊 T 15 Be part of competitions, organizations and societies that aim to preserve and defend natural resources including water. Speak to family and friends about donating or helping out charity groups to provide water to the most needy places.

Use water wisely

- 🔊 T 16 (4) Never assume that your society is too advanced to experience water shortage. If we do not acquire the right attitude towards water, it is only a matter of time and one day there will be a shortage. Keep the tap off when not in use. Minimize the flushing of toilets and bath times. In effect, anything that you can do to save water, do it.

Decide whether the following statements are true (T) or false (F), based on the article.

- ☐ **T** 1 Water is a precious natural resource. But many people don't seem to see that.
- ☐ **F** 2 Fresh water is surface water in the form of rivers, lakes and swamps.
- ☐ **T** 3 A global water crisis will come sooner or later if we do not learn to use water wisely.
- ☐ **F** 4 Water scarcity causes climate change directly.
- ☐ **T** 5 Low quality drinking water creates a health problem in many developing countries.

- T** 6 Agriculture needs far more water than any other economic sector.
- T** 7 Urbanization is a factor that causes water shortage.
- T** 8 Raising public awareness of the water problem is the first thing you can do toward its solution.
- F** 9 Desalination is not necessary because the 3% fresh water is enough for all humans and animals.
- T** 10 There is a high chance that people who have easy access to water do not think water is a problem.

Translate into Chinese the underlined sentences in the article.

🔑 1 Less than 3% of water on Earth is fresh water, and the bulk of this is trapped in snowfields and glaciers and not easily accessible.
 地球上的水只有不到3%是淡水，并且大部分都被困在雪地和冰川中，不容易获取。

🔑 2 ...but unfortunately, many factors have caused a major upset in the flow and use of fresh water and have caused massive crises in many regions of the Earth.
 ……但不幸的是，许多因素严重扰乱了淡水的流动和使用，使地球的许多地区出现大面积的危机。

 **3** This means if we do nothing in terms of preserving and using water wisely, it is only a matter of time before all regions begin to experience water crises and all the repercussions that come with it.

这意味着如果我们在保护水和明智用水方面无所作为，所有地区迟早都将遭遇水危机及随之而来的影响。

 **4** Never assume that your society is too advanced to experience water shortage.

绝不要以为你所处的社会先进，不至于会出现缺水问题。

Reading 2

🔊 China's Water Situation

Katie Burkhardt

🔊 T 1 Water quality is a major concern in China largely due to man-made pollution and natural **contamination**. According to a long-term study completed in 2011 by the Ministry of Environmental Protection and the Chinese Academy of Engineering, over 90 percent of the groundwater in cities was polluted to different degrees. Of 118 major cities, 64 percent had seriously contaminated groundwater supplies. **Furthermore**, laboratory tests **conducted** by the Ministry of Water Resources found that 20 percent of China's rivers were so badly polluted that their waters were too dangerous to even **come in contact with**. These alarming water



pollution statistics can be **traced** back to sources including manufacturing **emissions**, poorly treated sewage, industrial spills, and agricultural runoff. With 70 percent of China's population depending on groundwater for their drinking supply, these startling numbers present a very serious health concern.

 **T** 2 Unfortunately, pollution is only half of the country's water problem. Not only might China's water be contaminated; it is also **scarce**. The Ministry of Water Resources reported in 2012 that nearly two-thirds of China's cities are "water-needy" and more than 300 million **rural residents** lack access to drinking water. The Ministry **concluded** that the nation, **on the whole**, is only using 50 percent of its water supply **effectively**, with the agricultural industry held **accountable** for **significant** levels of wasted water.

 **T** 3 The good news is that China **displays** both the knowledge and the will to **rise to** the environmental and health challenges

its rapid development has created. To conserve its water resources, broad measures have been taken at different levels. The government has introduced efforts to **tighten** its supervision over **exploitation** of underground water, further protect sources of drinking water, and restore the aquatic **ecological** system. In agriculture, high-efficiency water-saving irrigation has been introduced. Meanwhile, advanced technologies have been adopted to increase the reuse rate of industrial waste water and reduce the total amount of water **consumed** in industries. Water supply systems in cities have been improved to minimize the risk of pipe damage, and domestic water saving and filtration apparatuses have been introduced.

🔊 **T** 4 The success of such sustainable water conservation efforts encourages **confidence** that, despite the seriousness of the problems, China has **at its command** the resources and resourcefulness needed to overcome them.

Skim the article above and answer the following questions.

🔑 1 What are the key words that may sum up China's water problem?

Pollution and scarcity.

🔑 2 What are the sources of water pollution?

The sources of pollution include manufacturing emissions, poorly treated sewage, industrial spills and agricultural runoff.

🔑 3 What has the Chinese government done for water conservation?

It has introduced efforts to tighten its supervision over exploitation of underground water, further protect sources of drinking water, and restore the aquatic ecological system. In agriculture, high-efficiency water-saving irrigation has been introduced. In industries, advanced technologies have been adopted to increase the reuse rate of industrial waste water and reduce the use of water. Water supply systems in cities have also been improved.

 4 Why is the author so sure that China will be able to overcome

its water problem?

She believes that China has both the knowledge and the will to effectively deal with the environmental and health challenges its rapid development has created. Success in its sustainable water conservation efforts encourages her confidence that China has both the resources and resourcefulness needed to address the problem.

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Harmony Between Man and Nature

“Harmony” is at the core of traditional Chinese culture and the Chinese people’s view of the world and the universe. Chinese people often say “heaven and man are an integral one,” and they are in constant pursuit of harmony between man and nature. The central authorities have now put forward the idea of creating a harmonious society. The harmony between man and nature often impacts on people-to-people relations and the relationship between individuals and the society.

Unfortunately, the connection between man and nature has been compromised in part due to man’s irrational exploitation of nature. A pleasant environment is what matters in the construction of a harmonious society, and protecting the environment is one of the main tasks we are facing now. China’s water situation suffers a lot currently, so what should we do?



Pair work: Work with your partner to complete the following dialogue, using the words and phrases in brackets. Make changes or additions where necessary.

A: Hello and welcome. Would you like a glass of water?

B: Yes, please. You see, offering someone a drink of water is always the polite thing to do. Water is of major importance to all living things. Up to 60 percent of the human adult body is water.

A: Yes. Human life and production activities are inseparable from water. However, with population growth, economic development, and climatic deterioration, the water situation is worsening for every continent.

B: I agree. China is also faced with serious water problems.

A: Now water quality is a major concern in China as a result of man-made pollution and natural 1) contamination. In some areas what was supposed to be the primary water source for humans was seriously polluted and became too dangerous to even 2) come in contact with. (come in contact with, contamination)

B: What could be the reasons for the alarming water pollution?

A: The pollution could be 3) traced back mainly to humans' improper activities in industrial and agricultural production, such as oil spills, harmful 4) emissions, poorly treated sewage. Considering 70 percent of China's population 5) depending on groundwater for drinking water, the situation of water pollution 6) presents a great health risk. (depend on, emission, present, trace back)

B: To make matters worse, water is not only contaminated, but also 7) scarce. Many rural 8) residents in China do not have access to drinking water yet. 9) On the whole, half of China's water supply is not used effectively, and the agricultural industry is considered to 10) be held accountable for much wasted water. (be held accountable for, on the whole, resident, scarce)

A: Don't worry too much. China can certainly 11) rise to those environmental and health challenges. In fact, a wide range of measures have been taken at different levels to 12) conserve water resources. (conserve, rise to)

B: That's good news. I see many successful initiatives have increased people's confidence. Water resources are part of nature. It's our duty to use water sustainably and rationally to contribute to the harmony between man and nature.

A: Sure thing. Starting with dealing with water problems, more environmental problems can have their long-term solutions, too. I believe a much more beautiful China with green mountains and clean waters will be here to impress the world. So, would you like another glass of water?

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1. Read the following passage carefully. Select one word for each blank from a list of choices given in the word bank. You may not use any of the words in the bank more than once.

access	conducts	consists	consume
contamination	drinking	emissions	furthermore
irrigate	polluting	resources	scarcer
sustainable	unfortunately	whole	

Water is a necessity for all life on Earth. Some even predict it to be the “new oil”, but why isn’t there enough for everyone? Even though our blue planet 1) consists of 70% water, we can only use an extremely small part of it as 2) drinking water. As much as 97% of all water is salty, and 2% is frozen. The remaining 1% could still be sufficient for everyone, but 3) unfortunately, its distribution is pretty unfair and the quality is inferior largely due to man-made pollution and natural 4) contamination.



access

conducts

~~consists~~

consume

~~contamination~~~~drinking~~

emissions

furthermore

irrigate

polluting

resources

scarcer

sustainable

~~unfortunately~~

whole

Today more than 700 million people have no 5) access to clean water. They have to walk miles to collect it and carry it home. In agriculture, groundwater and water from marshlands are used to 6) irrigate the fields in countries. With a dry and hot climate this is especially problematic. Once pumped out of the ground, the water doesn't come back. On the 7) whole, the water problem will become a major concern in many societies.

access	conducts	consists	consume
contamination	drinking	emissions	furthermore
irrigate	polluting	resources	scarcer
sustainable	unfortunately	whole	

The local population must live on dry land, and as the world's population grows, the demand for water is growing, too. The UN estimates that in 2025 demand will be up to 40% higher than it is today, and water will be 8) scarcer in some places. This bears the potential for conflicts about the use of water 9) resources. Governments and societies need to work out a(n) 10) sustainable and long-term solution. Technology must be upgraded, and new frameworks for intelligent water resource management have to be established, because there is no life without water.

2. Translate the following passage into English.



中国是全球缺水最严重的国家之一，约四分之一的省份面临严重缺水问题。随着城镇化人口增加以及污染加剧，用水需求不断增长，水资源短缺愈发严重。

China is one of the most water-scarce countries in the world, with about a quarter of its provinces facing severe water shortages. With the increase in the urbanized population and pollution, the demand for water is growing and the shortage of water resources is becoming more serious.

✦ 我国许多地区已出现因水资源短缺影响人民生活、制约经济发展的局面。针对水资源短缺的严峻态势，党和国家提出了“提高用水效率、全面推行各种节水技术和措施、建立节水型社会”的目标和任务。

In many areas of China, water shortage has affected people's lives and constrained economic development. In response to the severe situation of water shortage, the Party and the State have put forward the goal and task of "improving the efficiency of water use, comprehensively implementing various water-saving technologies and measures, and establishing a water-saving society."

At the beginning of this unit, we watched a part of the video clip about the coming global water crisis. Now we will continue to watch more to see how serious it is, not only in Asia, Africa, but also in America. After viewing the video clip twice complete the following sentences according to what you have heard. Before you start, getting to know the following words and expressions might be helpful.



Aral /'ærəl/ **Sea** （中亚）咸海

Ogallala Aquifer /əʊgə'lala 'ækwɪfə(r)/ （美）奥加拉拉含水层

Colorado Rockies /,kɒlə'rɑ:dəʊ 'rɒkɪz/ （美）科罗拉多州的落基山脉

South Dakota /də'kəʊtə/ （美）南达科达州

Texas Panhandle /'teksəs 'pæn,hændl/ （美）得克萨斯州最北部的一个地区

Vegas = Las Vegas /lɑ:s'veɪgəs/ （美）拉斯维加斯（内华达州著名赌城）

Conservation Reserve Program (CRP) 美国政府的一项土地休耕保护计划

1 Segment 1

Consider the Aral Sea, for example, located in central Asia. The name Aral Sea is actually translated into “Sea of Islands”, referring to over 1,534 islands that once existed. Today the Aral Sea is down to 10% of its original (最初的) size . The same thing is happening in other areas that were once flourishing (茂盛的) (but) are now turning into desert .



2 Segment 2

In the U.S., the world's largest body of fresh water, the Ogallala Aquifer, an underground lake that extends (延伸) from the Colorado Rockies to South Dakota going all the way down to the Texas Panhandle with a range (范围) of 50 to 300 feet of depth. This water resource is what has made America's plains (平原) the “ bread basket ” of the world. Unfortunately, like an oil well, this won't last forever . The Ogallala used to have an average depth of 240 feet. Today its average depth is just 80 feet .

3 Segment 3

What once was a booming (兴旺发达的) town that relied (依靠) on the Ogallala Aquifer, has now seen its depths of underground water fall between 0–50 feet. Many wells are completely dry and farmers have been forced to hand over their land to the government's Conservation Reserve Program (CRP) in exchange for grants (拨款).

On-the-spot (现场的) inquiry

Imagine yourself to be a reporter on your way to a village to learn about the quality of the drinking water there. You happen to see a villager in the village square and start to talk to him. Role-play the conversation with your partner. You may start your conversation as follows:

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Reporter: Excuse me, sir, do you live in this village?
Villager: Yes, I do. What can I do for you?
Reporter: ...
Villager: ...

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While preparing for the role-play, you are encouraged to use words and expressions you have learned from this unit, such as “source of the drinking water” and “affect people’s health”.

Tips



Work with your partner to find ways to conserve clean water supplies. Then write a paragraph about it. Make sure that the paragraph centers around a topic sentence.

Tips The topic sentence sets forth the main idea of the paragraph such as the following examples taken from the Text:

But water is complicated. (Para. 7)

Thankfully, there are solutions. (Para. 8)

When you finish, please exchange your writing with a classmate and evaluate each other's writing according to the following criteria:



Evaluation criteria:



- 1 You must write a paragraph of no less than 120 words.
- 2 The paragraph should begin with a topic sentence.
- 3 You should list a few things we can do to preserve clean water supplies.
- 4 You should end the paragraph by repeating the idea of the topic sentence, preferably in a different way.
- 5 You must use:
 - five key words and expressions from the Text,
 - two new words from the other two articles.Highlight these words and expressions clearly by underlining them or using a different color.

A Deeper Understanding of the New Development Concepts

harmonious /hɑ:'məʊniəs/

a. 和谐的; 和睦的

coexistence /,kəʊɪg'zɪstəns/

n. 共存; 和平共处

retaliate /rɪ'tæliet/

vi. 报复

drainage /'dreɪnɪdʒ/

n. 流域

Loess Plateau /'ləʊɪs 'plætəʊ/

黄土高原

sparse /spɑ:s/

a. crisscross /'krɪskrɒs/

vt. (在……内) 纵横交错

vegetation /,vedʒɪ'teɪʃn/

n. 植被; 草木

arable /'ærəbl/

a. 适于耕种的; 可开垦的

vegetation — were covered by dense forests and crisscrossed

by clean streams and natural pastures; however,

deforestation — severely damaged the local eco-

deforestation /di:'fɔ:ri'steɪʃn/

n. 毁林; 滥伐森林

pasture /'pɑ:stʃə(r)/

n. 草地; 牧场

session on implementing the decision of the Fifth Plenary Session of the 18th CPC Central Committee, attended by officials at the provincial/ministerial level.



the once-prosperous Silk Road. The expansion of the Hexi Corridor and the ancient city of Loulan. The ancient city of Loulan, situated on the Pecos River, changed from a lush meadow to a barren stretch of pinewoods, but the land reclamation policy during the 1862–1874 of the Qing Dynasty reduced the 500-kilometer stretch of pinewoods to barren mountains. We must take warning from these mountains.

T 3 Since reform and open-up, China's economic development has made historic progress. This is what we are proud of and also what has won the admiration of other countries. Nevertheless, we must be aware that ecological problems have become our country's a pressing public concern. For example, frequent incidences

reclaim /rɪ'kleɪm/
vt. 开垦; 改造

abound /ə'baʊnd/
vi. 大量存在; 充满

lush /lʌʃ/
a. 茂盛的

meadow /'medəʊ/
n. 草地; 牧场

reclamation /,rekle'meɪʃn/
n. 开垦

stretch /stretʃ/
n. 连绵的一片

pinewood /'paɪnwɒd/
n. 松木; 松林

barren /'bærən/
a. 贫瘠的

admiration /,ædmə'reɪʃn/
n. 钦佩; 赞赏; 羡慕

of environmental pollution have **endangered** lives and caused widespread **distress**. We must make every effort to change this situation.

endanger /ɪnˈdeɪndʒə(r)/
vt. 危及；使遭到危險

T4 Eco-environmental distress substitutes. We are not aware of this when we exploit it, yet it is **irretrievable** once lost. As I

distress /di'stres/
n. 痛苦; 不幸

previously mentioned **equate** /ɪ'kweɪt/ **live** **liveliness** **irretrievable** /,ɪrɪ'tri:vəbl/ **g.** 不能弥补的; 不能复原

equate /I'kweɪt/
vi. 等同

irretrievable /ˌɪrɪˈtriːvəbl/
a. 不能弥补的；不能复原的

Protecting the environment **equates** to protecting productive forces; improving the environment **adhere** to the fundamental state

livelihood /'laɪvliːhʊd/
n. 生计; 生活

policy of **conserving** resources. We should protect the environment. We should **adhere** to the ecological principle like our own eyes, **vi** 坚持, 遵守

adhere /əd'hɪə(r)/
vi. 坚持; 遵守

and to protect the environment as our own lives. We should
develop a green way of development and a green way of life,

conserve /kən'sɜ:v/
vt. 保护

as a support to increasing individual and national prosperity
and creating a beautiful environment.

T 5 We should ensure the people breathe fresh air, drink clean water, eat safe food, and enjoy a **livable** environment. We should ensure the people receive the real environmental benefits from economic development, green mountains, clear waters, and a better environment than ever before, and step into a new era of eco-progress.

livable /'lɪvəbl/

a. 适于居住的；宜居的

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Match the English expressions in the left column with their corresponding Chinese versions in the right column. Write down the letter before each number.

g **1** harmonious coexistence of humanity and nature

e **2** respect, accommodate, and protect nature

a **3** have endangered lives and caused widespread distress

c **4** protecting productive forces

b **5** adhere to the fundamental state policy

d **6** develop a green mode of development and a green way of life

h **7** receive the real environmental benefits from economic development

f **8** step into a new era of eco-progress

a) 成为民生之患、民心之痛

b) 坚持基本国策

c) 保护生产力

d) 形成绿色发展方式和生活方式

e) 尊重自然、顺应自然、保护自然

f) 走向生态文明新时代

g) 人与自然和谐共生

h) 感受到经济发展带来的实实在在的环境效益

Translate the following sentences into Chinese.

- 🔑 **1** Human development activities must respect, accommodate, and protect nature; otherwise nature will retaliate against us.

人类发展活动必须尊重自然、顺应自然、保护自然，否则就会遭到大自然的报复。

- 🔑 **2** Environment is livelihood, green mountains are beauty, blue sky is happiness, and clear water is wealth.

环境就是民生，青山就是美丽，蓝天就是幸福，绿水青山就是金山银山。

- 🔑 **3** Protecting the environment equates to protecting productive forces; improving the environment equates to developing productive forces.

保护环境就是保护生产力，改善环境就是发展生产力。

🔑 4 We should protect the eco-environment like our own eyes, and treat the eco-environment as our own lives.

我们要像保护眼睛一样保护生态环境，像对待生命一样对待生态环境。

🔑 5 We should ensure the people breathe fresh air, drink clean water, eat safe food, and enjoy a livable environment.

我们要让老百姓呼吸上新鲜的空气、喝上干净的水、吃上放心的食物、生活在宜居的环境中。

Words and Expressions

faucet /'fɔ:sɪt/ *n.* 水龙头

swamp /swɒmp/ *n.* 沼泽；湿地

germ /dʒɜ:m/ *n.* 细菌

diarrhea /,daɪə'riə/ *n.* 腹泻；痢疾

dehydration /,di:'haɪ'dreɪʃn/ *n.* 脱水；极度口渴

solvable /'sɒlvəbl/ *a.* 可以解决的

brand-new /'brænd'nju:/ *a.* 崭新的；新制的

age-old /'eɪdʒəʊld/ *a.* 古老的；由来已久的

attendance /ə'tendəns/ *n.* 出席；到场

Watch the video clip “Water Changes Everything.” Complete the following statements with what you’ve heard.  

The water crisis:

- Nearly 1 billion people live without clean 1) drinking water.
- They don’t have a faucet. Instead the women and children go off to 2) collect water.
- Some women spend more time collecting water than any other activity they do in the day, and the walk isn’t just hard; it’s 3) dangerous. Many get hurt; sometimes they’re even 4) attacked.
- Their water is contaminated with germs that cause diarrhea, dehydration, even death, but what 5) choice do they have?

The solutions:

- Some solutions are brand-new and innovative, like water filtration 6) systems; some are age-old, like drilled or hand dug 7) wells.

Significance of the solutions:

- These solutions bring clean water 8) much closer to the people who need it.
- Africa alone could save 9) 40 billion hours each year.
- That's less money spent on 10) medicine, which means more money for books and school 11) uniforms.
- A water project built near a school can 12) increase attendance, especially among young girls.

1 天地与我并生，而万物与我为一。（战国 庄子《齐物论》）
The universe and I came into being together; I and everything **therein** are One.

2 人法地，地法天，天法道，道法自然。（春秋 老子《道德经》）
therein /ˈðeər'in/
ad. 在其中；在那里

Man models himself after the Earth;
The Earth models itself after Heaven;
The Heaven models itself after Tao,
Tao models itself after nature.

3 采菊东篱下，悠然见南山。山气日夕佳，飞鸟相与还。（东晋 陶渊明《饮酒·其五》）
aster /'æstə(r)/
n. 紫菀；翠菊

I pick fence-side **asters at will**;
Carefree I see the south mountain night;
The mountain air's fresh and sweet;
Together birds go home in flight.
at will
随意；任意

4 美的真谛应该是和谐。这种和谐体现在人身上，就造就了人的美；表现在物上，就造就了物的美；融汇在环境中，就造就了环境的美。（冰心）

The true meaning of beauty should be harmony. When it is **manifested** in a person, it creates human beauty; when it is manifested in an object, it creates the beauty of the object; when it is manifested in the environment, it creates the beauty of the environment.

manifest /'mænɪfɛst/
vt. 表明，显示

5 人要学会征服。征服，不在于力的征服，而在于学会征服。（钱钢 《唐山大地震》）

conquest /'kɒŋkwɛst/
n. 征服；战胜

The final **"conquest"** of nature by man does not lie in the conquest of force, but in learning to live in harmony with nature.

Group project: The water problems are a major concern in China. China's water resources are unevenly distributed. Over 90% of the groundwater in cities is contaminated to different degrees, and demand for water has been increasing. What policies have been made by the government to solve the water problems? What responsibilities do you think citizens should take to help address these problems? Work in groups and discuss the questions. Each group selects one person to share your discussion result with the class.

Upon completion of this unit, you are expected to:

- have a thorough understanding of the text linguistically and contextually;
- build up a basic vocabulary about humanity's water problem and learn to use the key words and expressions;
- discuss the water situation in the world and in China;
- explain why water is going to be the greatest challenge facing mankind in the 21st century;
- propose ways to help conserve water;
- write a passage centering around a topic sentence about water conservation.

Tips for your reference

1. Decide the topic.

1.1 For this task, the topic is “Ways to conserve clean water supplies”.

1.2 Write a topic statement.

(e.g. In the face of the global water crisis, we should take action to conserve clean water supplies ...)

2. Arrange the structure and then write an outline.

2.1 This topic may include the following supporting ideas:

— Why should we conserve clean water supplies?

(e.g. The source of fresh water is limited and fresh water is seriously polluted in recent years.)

— What can we do to protect clean water supplies?
(e.g. *classify garbage, prohibit pouring polluted water into rivers or seas, develop the sewage system ...*)

2.2 Choose proper linking words to connect different ideas:
(e.g. *first ... then... later... in the end ...; because/since/as; on one hand ... on the other hand ...*)

3. Check for spelling mistakes and grammar errors.

水的问题怎么办

迈克尔·E·韦伯

2007年，我女儿七岁。每晚我们一起刷牙，这是我们每天必守的老规矩。为了节水，我们先把牙刷湿一下就关上龙头，只到漱口时才再打开龙头。一天晚上，女儿认为我关龙头动作不够快，不乐意了。她关上龙头，对我做了个生气的鬼脸，嘟哝着说：“爸，关上龙头，科学家需要时间。” (Para.1)

这句话至今犹言在耳，这不仅是因为她小小年纪就那么懂事，也因为她完全正确。我们必须节约资源，争取时间，使科学家来得及找到解决问题的新方法。对水而言更是如此。

(Para.2)

水有望成为21世纪最重要、最可能引起争议的资源。它有可能取代石油，成为触发地缘政治冲突的战略资源。然而，如果有正确的解决方法，它也可能成为使我们团结一致的资源。

(Para.3)

2003年，已故的诺贝尔奖得主理查德·斯莫利在莱斯大学举办的能源和纳米技术会议上曾做过一个讲座，着重谈了未来五十年人类面临的十大问题。他列出的问题清单，按重要性排序，依次是：能源、水、食品、环境、贫困、恐怖主义和战争、疾病、教育、民主，以及人口。（Para.4）

能源和水位居榜首，排在食品和贫困之前，其原因是解决前两个问题，后面的问题就比较容易解决了。开发充足的、清洁的、可靠的、用得起的能源，就能保证有充分的净水供应。有充分的净水就能生产食品，并保护环境，以此类推。（Para.5）

从许多方面讲，20世纪是能源冲突的世纪。现在，随着太阳能的价格下降、分散发电即将大范围推广，能效即将大大提高，新能源时代的黎明即将到来。我们可以预见，本世纪某个时候，水将取代能源，成为人类下一个巨大挑战。恰当解决水资源问题，能为我们走向一个彻底解放的、健康与和平的文明扫除障碍。

(Para.6)

但水的问题很复杂。首先，它没有替代物。煤和油都有替代物，但水不能被取代。第二，水和社会的每个方面都密切相关，能源生产要靠水来冷却发电厂、压裂油页岩，农业要水灌溉，工业和城市因各种用途要消耗数万亿加仑的水。第三，水的需求量在增长，而供应量却上下波动。气候变化会加剧旱灾和水灾，并改变降水的时间和地点。这意味着人类将面临抉择，是让人移居到有水的地方，还是把水引向有人的地方。

(Para.7)

谢天谢地，水问题有办法解决。有的办法规模宏大，资金和能量的耗费高得难以置信，如建造跨洲的运水渠、新的水电大坝、大型海水净化厂。有些办法是小型的，如用微型集水器把空气中的水汽凝集成水。所有这些办法的实际应用都还需要一段时间。

(Para.8)

1961年4月，约翰·F·肯尼迪总统说：“如果我们能有竞争力地——也就是以低廉的价格——把海水净化成淡水，这将有利于人类的长期利益，这一成就将使其他所有科学成就都显得渺小。”这是他在著名的登月演说前一个月说的话。他讲话后不到十年，我们把人送到月球并成功返回，但我们仍未找到一个办法，可以不花很多钱和能源就把海水变成淡水。这主要是因为并未将宏大的创新机制用到水上。科研预算优先能源、国防和健康领域的项目，而往往忽略了水，尽管解决水问题有助于解决能源、国防和健康方面的问题。在火星上找水我们花了很多钱，如果用同样的钱在地球上找水，结果就会大不相同。

(Para.9)

我们最终有能力解决水问题。但我们需要积极地节水，这会帮我们赢得时间，让发明家寻找方法。 (Para.10)

全球缺水

对很多人来说，水从来就不是生活中的大事，这是因为他们生活在有良好供水系统的社区里。一年到头，他们一开龙头就有净水流出，这使人们很难意识到水有多宝贵。（Para.1）

大学资料都在这里

水是生命，动植物和人都靠这个无价的自然资源生存。除此之外，水还被用于清洁卫生、制造业、建筑业和农业。你能想到的几乎每一种人类活动都多少涉及用水。（Para.2）

地球表面百分之七十以上被水覆盖，怎么还会缺水？

地球上的水只有不到3%是淡水，并且大部分都被困在雪地和冰川中，不容易获取。其余的水形成了海洋，不能像淡水一样使用。只有很小一部分（0.014%）是以河流、湖泊和湿地形式存在的地表水。

（Para.3）

自然，这3%的淡水足以让地球上的所有人和动物生存，但不幸的是，许多因素严重扰乱了淡水的流动和使用，使地球的许多地区出现大面积的危机。(Para.4)

但你为何要关心此事？

我们应该关心此事，因为造成缺水的很多因素，如人口增长、城市化、污染和气候变化，都在扩大，变得更复杂和更不可控。这意味着如果我们在保护水和明智用水方面无所作为，所有地区迟早都将遭遇水危机及随之而来的影响。（Para.5）

缺水的后果有哪些？

缺水的后果可归为四大方面，即健康、饥饿、教育和贫困。
(Para.6)

大学资料都在公众号：一号教学楼

健康

在许多发展中国家，人们只能饮用来自溪流的劣质水，这些溪流很多是被污染的。有很多通过水传播的疾病致人死命。

(Para.7)

大学资料都在公众号：三哥说英语

缺水或缺优质水导致很大的卫生问题。诊所、餐馆、公共厕所和许多其他地方清洗时不得不用很少的水。这就增加了管理和使用这些设施的人的健康风险。 (Para.8)

饥饿

种植食物、饲养动物需要很多水。专家说，全球而言，我们用于农业和灌溉的水占我们水资源的70%。家庭用水只占10%。
(Para.9)

大学资料都在公众号：大学资料网

水少了，靠水生长的农作物产量便降低；没有水，牲口会死，其他动物也不会好。结果是不间断的饥渴和低质量的生活。

(Para.10)

教育

水和教育的相关不是一下子就能看明白的。在世界许多地方，很多儿童天一亮就得起床为全家取水。他们要走好几英里地才能取到水。孩子们很累，有的因此而缺课。（Para.11）

贫困

能否获得优质水是繁荣经济和提高生活水平的关键。制造业、种植业、采矿业都需要大量的水才能发展。缺水就意味着经济活动停止，人们只能生活在永久贫困之中。（Para.12）

你在保护水资源中能起什么作用？

有时问题太大会使人感到无能为力。但是你能为水做不少事。很可能读到这篇文章的人并不生活在缺水地区，因此会认为这不是他们的问题。下面就是你能为节水可做的事。（Para.13）

认识

获取有关水危机的知识，就像你现在正在做的那样。你如果了解了问题，就能更好地解决问题。与家人和朋友谈谈这个问题。关注缺水问题和危机地区的新闻和情况。（Para.14）

参与

参加旨在保护包括水在内的自然资源的各种竞赛、组织和社团。与家人和朋友谈谈，看他们能否捐款或帮助为最缺水的地方供水的慈善机构。

(Para.15)

明智用水

绝不要以为你所处的社会先进，不至于会出现缺水问题。如果我们对水的态度不正确，总有一天会缺水，只是时间问题。不用水时关上龙头，尽量减少马桶冲水和洗澡次数。就是说，只要你能做的可以节水的事，你就要去做。 (Para.16)

中国的水况

凯蒂·伯克哈特

在中国，水质是一个主要的关注点，这主要是因为人为的和自然的污染。中国环保部和中国工程院于**2011**年完成的一项长期研究表明，**90%**以上的城市地下水有不同程度的污染。**118**个大城市中，**64%**的城市的地下水源严重污染。更有甚者，水利部进行的检测发现，中国**20%**的河流受到严重污染，即使仅仅与这些河水接触都非常危险。这些惊人的水污染数据可以追溯到其污染源：制造业的排放，污水处理不力，工业废水外溢，以及农业径流。由于中国**70%**的人口的饮用水是地下水，这些惊人的数据显示了一个非常严重的健康隐忧。（Para.1）

不幸的是，污染只是中国水问题的一半。中国的水不仅可能受污染，而且还很稀缺。2012年水利部报告称三分之二的城市缺水，三亿多农村居民得不到饮用水。水利部因此得出结论，总体上，中国对供水的使用只有50%是有效的，而农业则是造成水资源惊人浪费的大户。

(Para.2)

好消息是，中国表现出有学识、有意愿，能够很好应对其快速发展引起的环境与健康方面的挑战。为保护水资源，中国已在不同层面采取了广泛的措施。政府已加强对地下水开采的监管，进一步保护饮用水源，恢复水生态系统。在农业领域，已引进高效节水灌溉技术。与此同时，采用先进技术，提高工业废水的回用率，减少工业用水总量。城市供水系统已得到改善，最大限度地减少管道损坏的风险，此外，家用节水和过滤设备也被采用。

(Para.3)

这些可持续的水资源保护工作的成功鼓舞了人们的信心，
尽管问题严峻，但中国拥有解决这些问题所需的资源和睿智。

(Para.4)

大学资料都在公众号：三号教室

深入理解新发展理念

2016年1月18日

绿色发展，就其要义来讲，是要解决好人与自然和谐共生问题。人类发展活动必须尊重自然、顺应自然、保护自然，否则就会遭到大自然的报复，这个规律谁也无法抗拒。（Para.1）

*这是习近平在省部级主要领导干部学习贯彻党的十八届五中全会精神专题研讨班上的讲话。

据史料记载，现在植被稀少的黄土高原、渭河流域、太行山脉也曾是森林遍布、山清水秀，地宜耕植、水草便畜。由于毁林开荒、滥砍乱伐，这些地方生态环境遭到严重破坏。塔克拉玛干沙漠的蔓延，湮没了盛极一时的丝绸之路。河西走廊沙漠的扩展，毁坏了敦煌古城。楼兰古城因屯垦开荒、盲目灌溉，导致孔雀河改道而衰落。河北北部的围场，早年树海茫茫、水草丰美，但从同治年间开围放垦，致使千里松林几乎荡然无存，出现了几十万亩的荒山秃岭。这些深刻教训，我们一定要认真吸取。

(Para.2)

改革开放以来，我国经济发展取得历史性成就，这是值得我们自豪和骄傲的，也是世界上很多国家羡慕我们的地方。同时必须看到，我们也积累了大量生态环境问题，成为明显的短板，成为人民群众反映强烈的突出问题。比如，各类环境污染呈高发态势，成为民生之患、民心之痛。这样的状况，必须下大气力扭转。

(Para.3)

生态环境没有替代品，用之不觉，失之难存。我讲过，环境就是民生，青山就是美丽，蓝天也是幸福，绿水青山就是金山银山；保护环境就是保护生产力，改善环境就是发展生产力。我们要坚持节约资源和保护环境的基本国策，像保护眼睛一样保护生态环境，像对待生命一样对待生态环境，推动形成绿色发展方式和生活方式，协同推进人民富裕、国家强盛、中国美丽。

(Para.4)

我们要让老百姓呼吸上新鲜的空气、喝上干净的水、吃上放心的食物、生活在宜居的环境中、切实感受到经济发展带来的实实在在的环境效益，让中华大地天更蓝、山更绿、水更清、环境更优美，走向生态文明新时代。（Para.5）

 conserve

vt. use without wasting; protect from loss or destruction
节约使用；保护

 To **conserve** energy, most families in this apartment building do not turn on the air-conditioners except in extreme weather.

为节约能源，这幢楼大多数人家若非极端天气一般不开空调。

conservation

n. 节约使用；保护

 Water **conservation** tops the list of our urgent tasks.
节水是我们工作的重中之重。

大学资料都在公众号：三号教学楼

to one's liking

中意，称心

 “We do our best to ensure that the food we serve is always **to your liking**,” said the owner of the restaurant to us.

“我们尽力提供你们满意的菜肴。”饭店老板对我们说。

 I find local cuisine very much **to my liking**.

我发现当地的饮食很合我的口味。

 ring in one's ears

make one feel that one can still hear sth. 犹言在耳

 She went to bed with the song still **ringing in her ears**.
她上床睡觉，歌声仍萦绕在耳。

 My father's last word is still **ringing in my ears**.
父亲的临终遗言仍在我耳边回响。

resource

n. (usu. pl.) 资源

 Land **resources** in this small island country are extremely scarce.

这个弹丸岛国的土地资源极为稀少。

 Today we are overpopulated, straining the Earth's **resources**.

现在我们人口过多，致使地球资源使用过度。

 on track

likely to achieve the result you want 步入……轨道；
有望

 Please get the discussion **on track**. Time is limited.
请不要岔开话题。时间有限。

 The boy is **on track** to become a chess master.
这孩子有望成为象棋大师。

 replace

vt. be used instead of sb./sth. else, put sb./sth. new in the place or position of sb./sth. 取代

 Will computers ever completely **replace** human beings?

电脑会不会完全取代人类?

 An American college graduate has been hired to **replace** our previous English teacher.

我们聘用了一位美国大学毕业生来接替我们以前的英语老师。

 solution*n.* 解决方案

 We need to conserve our resources to buy ourselves time so scientists can find new **solutions** to our problems.

我们必须节约资源，争取时间，使科学家来得及找到解决问题的新方法。

 conflict

n. a fight, battle or struggle, a state of opposition between ideas, interests, etc. 冲突, 矛盾

 There is a long history of **conflicts** over water resources between the two countries.
两国长期存在水资源冲突。

 She found herself in **conflict** with her parents over her future career.
她发现自己在未来择业问题上与父母存在着分歧。

 highlight

vt. give special attention to (sth.); emphasize 使突出; 强调

 The teacher said: “Write a paragraph centering around a topic sentence, and **highlight** your topic sentence in yellow.”

老师说：“围绕中心句写一个段落，中心句用黄色标出。”

 The report **highlights** the major problems facing society today.

报告特别强调了当今社会所面临的主要问题。

 highlight

n. sth. very interesting, exciting or important, the best part of sth. 主要部分，提要

 I missed the game, but I saw the **highlights** on the evening news.

我错过了比赛，不过在晚间新闻看了精彩回顾。

 The **highlight** of our tour was seeing the palace.

我们旅途中最有意思的活动就是参观宫殿。

environment

n. 环境

 How can we create a workplace **environment** that encourages innovation?

我们如何营造鼓励创新的工作环境？

 Because of the extreme cold, the Antarctic is a uniquely fragile **environment**.

极度严寒造成了南极地区绝无仅有的脆弱环境。

 subsequent

a. happening after sth. else 随后的，后来的

 The house was built in 1980, and has been expanded several times in **subsequent** years.

这幢房建于1980年，之后数次扩建。

 In **subsequent** interviews the candidate denied what he had said that night.

在随后的采访中，这位候选人否认了他当晚所说的话。

abundant

a. 大量的，充裕的

 The project aims at providing **abundant** energy in power-starved areas of the world.

该项目旨在为世界上能源不足的地区提供充足的能源。

 Shandong province is famous for its **abundant** agricultural resources.

山东省以其丰富的农业资源而闻名遐迩。

 source

n. 来源；源头（指河流的源头，或资金、信息等的来源，不同于“resource”，后者常指体现国家、组织或个人实力的资源，如物力、财力、劳力以及土地、矿山等自然资源）

 Exhibitions are an important **source** of income for many artists.

开展会是不少艺术家重要的收入来源。

 Novels have been a rich **source** of material for the film industry.

小说一直是电影业丰富的素材来源。

reliable

a. 可靠的，可信赖的

 Dogs, if treated properly and trained with firm kindness, are faithful and **reliable**.

只要你对它好，训练时态度坚决而友善，狗都能变得忠实可靠。

 We are looking for someone who is **reliable** and hard-working.

我们在物色可靠而又勤奋的人。

enable

v. 使成为可能；使能够

 The conference **enabled** greater international co-operation.

这次会议加强了国际合作。

 The cell phone will **enable** you to call for help from anywhere.

手机可以让你在任何地方打电话求助。

and so forth

et cetera 等等，诸如此类（常用于表达“还有其他与上文同类的活动（事件、东西），不再逐一列出”）

 We shopped, had lunch, shopped some more, **and so forth**.

我们买买买，吃午饭，再买买买，就这样。

 He has done many jobs: typist, salesman, deliverer **and so forth**.

他做过很多种工作：打字员、推销员、快递员等等。

around the corner

not far away, coming very soon 不远；即将到来

 The petrol station is around the corner.
加油站就在拐角处。

 Spring Festival is just around the corner.
春节即将来临。

大学资料都在公众号：大学资料

go down

下降

 If housing prices continue to **go down**, many small banks will go bankrupt.

如果房价持续走低，不少小银行就会破产。

 She believes that interest rates will **go down** in the next few weeks.

她相信未来几周利率会下降。

 distribute

v. 分发；散布；分配

 As freshmen, they must learn quickly how to **distribute** their time among the several courses they take.
作为新生，他们必须很快学会不同的课程怎么分配时间。

 They believe that to **distribute** wealth properly is at least as important as to create wealth.
他们认为，正确分配财富至少与创造财富同等重要。

efficiency

n. 效率

 She is always seeking to improve the **efficiency** and effectiveness of teaching.

她一直在努力提高教学的效率和效果。

 A market economy is all about **efficiency**, not morality.
市场经济注重的是效率而不是道德。

on the point of

about to (do sth.) 正要……

 The researchers believe that they are **on the point of** a breakthrough.

那些研究者认为他们即将取得突破性进展。

 He was **on the point of** saying something when the phone rang.

他正要说话时电话铃响了。

 foresee

v. (foresaw, foreseen) think or know sth. is going to happen in the future 预见；预知

 Did scientists **foresee** the revolution in communication technology?

当时科学家有没有预见到这一通讯科技的重大变革？

 No one can **foresee** what will happen in the future.
谁也无法预见将来会发生什么事。

civilization

n. 文明

 Athens (雅典) is generally considered as the birthplace of Western **civilization**.

雅典被认为是西方文明的发源地。

 The Chinese **civilization** is one of the oldest in the world.

中华文明是世界上最古老的文明之一。

 complicated*a.* 复杂的

 Our physics teacher is very good at explaining **complicated** concepts in simple language.

我们的物理老师善于深入浅出讲解复杂的概念。

 This is the most **complicated** case I have ever handled.
这是我处理过的最为复杂的案子。

 alternative

n. 选择；供选择的東西

 Biking is a more environment-friendly **alternative** to car driving.

相对于开车，骑自行车更为环保。

 There is a vegetarian **alternative** on the menu every day.

每天的菜单上另有素食餐点可选。

intensify

v. increase in degree or strength 加剧，强化

 The police **intensified** their search for the escaped prisoner.

警方加强了对逃犯的搜捕。

 A hurricane could also **intensify** the ecological damage onshore.

飓风同样能够加剧对海岸生态的破坏。

be confronted with

meet face-to-face with 面对

 When **confronted with** the witnesses of the murder, the accused was speechless.

面对谋杀目击者，被告哑口无言。

 He **was confronted with** his childhood trauma and tried to exorcise the pain.

他正视自己儿时所受的创伤，努力消除心中的痛苦。

rate

n. 价格，比率

 The postal **rates** vary from state to state.
不同的国家邮资不同。

 The mortality **rate** of cancer has been going down in recent years.
近年来，癌症的死亡率一直在下降。

 rate

vt. place in a particular rank or grade 将……列入（某一特定的阶层或等级）

 She is **rated** among the top ten heart surgeons in the country.

她名列全国十大心脏外科医生。

 She is highly **rated** as a novelist.

她这位小说家受到很高的评价。

in the interest of

to the advantage of 对……有利，为了

 **In the interest of** the public, we do not allow dumping into the rivers.

为了公众的利益，我们不允许将垃圾倾倒入河。

 All I have done is **in the interest of** the family.

我做的一切都是为了家族利益。

 dwarf

vt. 使（显得）矮小；变矮小

 At over six feet in height, the twelve-year-old boy **dwarfs** his peers.

这个12岁的孩子身高超过6英尺，明显高于同龄孩子。

 Our little dinghy was **dwarfed** by the big yacht.

我们的小舢板跟大游艇一比显得很小说。

 dwarf

n. an abnormally small person 侏儒

 I remember mother reading to me the story of *Snow White and the Seven Dwarfs*.

我还记得妈妈给我读《白雪公主和七个小矮人》。

 The dwarf's long arms were not proportional to his height.

那侏儒的长臂与他的身高不成比例。

 innovation

n. 革新，创新

 In aiming for **innovation**, surround yourself with people that think differently from you.

要有所创新，就要和想法和你不同的人在一起。

 This technical **innovation** will save us much time and labor.

这项技术革新可以为我们节省大量的时间和劳力。

budget

n. 预算

 Is there any money left in our monthly **budget**?
我们每月预算里还有钱剩下吗？

 The project is well over **budget**.
这一项目远超预算。

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mostly

ad. 主要地；通常；大部分

 We have some cold days. But **mostly** it is warm.
天气有时会挺冷，但基本上挺暖和。

 Recycled materials are **mostly** of limited use.
再生材料大多用途有限。

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 outcome

n. 结果

 It was midnight on election day. Many people were still up waiting for the **outcome**.

选举日，午夜。许多人还没睡，在等待选举结果。

 The **outcome** of the election is a foregone conclusion.
这次的选举结果已在预料之中。

contamination

n. 污染

 Changing weather patterns may put some densely populated areas at risk of major flooding, along with the resulting **contamination** of food and water supplies.
天气模式的改变会让一些人口稠密的地区面临特大洪灾的风险，随之而来的是食物和水源的污染。

 Transport of seafood imports over long distances presents opportunities for **contamination** due to improper handling and refrigeration.
进口海鲜在长途运输中由于处理和冷藏不当，有可能受到污染。

 furthermore

ad. 此外，而且

 **Furthermore**, university students in Shanghai have a better chance to study abroad as exchange students.
而且，上海的大学生作为交换生出国留学的机会更大。

 **Furthermore**, China takes the people's well-being and common prosperity as its ultimate goal.
此外，我国以增进人民福祉、实现共同富裕为最终目标。

 conduct

vt. 实施

 By adding 40 medical centers to **conduct** the tests, Wuhan had 63 institutions analyze the test samples.
武汉又有40个医疗中心参与检测，这样全市就有63家机构对检测样本进行分析。

 The WHO is ready to further **conduct** strategic cooperation with China.
世界卫生组织愿同中方继续开展战略合作。

come in contact with

接触

 University gives every student an opportunity to **come in contact with** a vaster world and experience more new things.

大学给了每个学生一个机会，让他们接触到更广阔的世界，体验更多新鲜事物。

 The winner said, “I draw inspiration from everything I read and everyone I **come in contact with**.”

这位优胜者说：“我从读过的一切、接触过的每一个人身上获得灵感。”

 trace

vt. 追踪；跟踪

 A group of researchers reveal scores of studies that **trace** the relationship between health and mood.

一组研究人员揭示了大量追踪健康和情绪之间关系的研究。

 Today we are witnessing a sixth mass extinction, but unlike the others, the current loss of biodiversity can be **traced** to human activity.

今天，我们正在见证第六次物种大灭绝，但与以往几次不同的是，目前生物多样性的丧失可以追溯到人类活动。

 emission

n. 排放（物），散发（物）

 By 2025 the city will lower carbon dioxide **emissions** by 40% to 45% from the 2005 level.

到2025年，该市的二氧化碳排放量将比2005年降低40%至45%。

 We will accelerate the promotion and application of advanced energy-saving and carbon **emission** reduction technologies, encourage green consumption, and promote green and low-carbon ways of production and life.

我们将加快节能降碳先进技术的推广应用，倡导绿色消费，推动形成绿色低碳的生产方式和生活方式。

 scarce

a. 缺乏的；稀少的

 Despite its great popularity, research on tai chi is **scarce**.
尽管太极拳广受欢迎，但与之有关的研究少之又少。

 We should find better ways to make good use of **scarce** resources.

我们应该找到更好的方法来充分利用稀缺资源。

rural

a. 农村的

 A social security system covering both urban and rural residents has taken shape.

覆盖城乡居民的社会保障体系基本建立。

 All-round progress has been made in education, with remarkable advances made in the central and western and in rural areas.

教育事业全面发展，中西部和农村教育明显加强。

 resident

n. 居民

 Macao's return to the motherland is of great significance to every Macao **resident**.

澳门回归祖国，对每一位澳门居民而言都意义非凡。

 The representative called on the government to build more low-cost homes for local **residents**.

这位代表呼吁政府为当地居民建造更多廉价住宅。

 conclude

vt. 推断出

 The researchers **concluded** that the employees' sense of happiness is probably due to a better work-life balance.

研究人员推断，员工们的幸福感可能是源于在工作与生活中达到了更好的平衡。

 The experts **concluded** that almost one in every two jobs has a risk of being replaced by machines.

专家们得出结论，几乎每两个工作岗位中就有一个有被机器取代的风险。

on the whole

大体上，基本上

 Development in Sino-Japanese relations has **on the whole** maintained a positive trend.

中日关系的发展总体上保持了良好的态势。

 **On the whole**, the people have reached a well-off standard of living.

总的来说，人民生活已经达到小康水平。

 effectively

ad. 有效地

 We should improve the global investment rules and use development resources more **effectively**.

我们要完善全球投资规则，更加有效地配置发展资源。

 To help the flood victims more **effectively**, major charities are working together in the distribution of food and clothes.

为了更有效地帮助洪水灾民，各大慈善机构正在合力分发食品衣物。

 accountable*a.* 应负责的

 When you forgive someone, this does not mean that they are no longer **accountable** for their actions.

当你原谅他人的时候，并不意味着他们不再为自己的行为负责。

 Officials who fail to perform their duties should be held **accountable**.

对不履行职责的官员要问责。

significant

a. 相当数量的

 Since the turn of this century, a **significant** number of emerging markets and developing countries have achieved rapid economic growth.

新世纪以来，一大批新兴市场国家和发展中国家的经济快速发展。

 There are no **significant** differences between the two groups of students in terms of their IQ.

这两组学生在智商方面没有显著差异。

 display

vt. 显示；显露；表现

 The athletes **displayed** great willpower throughout the contest.

运动员们在整场比赛中都表现出了顽强的毅力。

 As a historical city where Chinese and Western cultures meet, Macao displays a unique charm that belongs to a culture typical of southern China, yet with a **distinctive** European touch.

澳门作为中西文化荟萃的历史文化名城，传承岭南色彩的中华文化，融汇欧陆风情，蕴含独特魅力。

 rise to

能够成功处理（困难局面等）

 Faced with increasingly fierce domestic and international market competition, the leaders should **rise to** the difficulties and lead the employees in breaking new ground.

面对日趋激烈的国内外市场竞争，领导者应该迎难而上，带领员工开拓进取。

 We should join hands and **rise to** the challenge and march towards a bright future.

我们应该携手迎接挑战，迈向美好的未来。

 tighten

vt. 加强，加紧

 The environment ministers agreed to **tighten** controls on the transport and disposal of waste.

环境部长们同意加强对废物运输和处置的控制。

 To enable humanity to benefit from knowledge and innovation, we need to **tighten** the protection of intellectual property.

为了让人类从知识和创新中受益，我们需要加强知识产权保护。

exploitation

n. 使用，利用

 To control the **exploitation** of natural resources and restore the ecosystem, we must follow the laws of nature.

控制自然资源开发，恢复生态系统，必须遵循自然规律。

 After long-term development, the **exploitation** of arable land in China is too intense.

经过长期发展，我国耕地开发利用强度过大。

 ecological*a.* 生态的

 The author shares her own vision story, which is one of human beings saving the planet from **ecological** disaster by working together.

作者分享了她的愿景故事——人类共同努力拯救地球免于生态灾难的故事。

 Economic development should not be achieved at the expense of the **ecological** environment.

经济发展不能以破坏生态为代价。

consume

vt. 使用；消费

 Some of the most efficient refrigerators **consume** 70 percent less electricity than traditional models.
一些能效最高的冰箱比传统冰箱耗电量少70%。

 How does what you **do**, what you **consume**, and how you live, impact the environment?
你做的事情、你消费的东西、你生活的方式，如何影响着环境？

 confidence

n. 信心

 The people of all ethnic groups in China should have full **confidence** in the path, theory, system and culture of socialism with Chinese characteristics.

全国各族人民一定要增强对中国特色社会主义的道路自信、理论自信、制度自信和文化自信。

 confidence

n. 信心

 Today, we are closer than at any time in history to attaining the goal of the rejuvenation of the Chinese nation, and we have greater **confidence** in and capability for achieving this goal than ever before.

现在，我们比历史上任何时期都更接近实现中华民族伟大复兴的目标，比历史上任何时期都更有信心、更有能力实现这个目标。

at (one's) command

可自由使用（或支配）

 With all these resources **at our command**, we can accomplish the project in no time.

凭借我们手头的所有这些资源，我们可以在很短时间内完成这个项目。

 The local army men helped the victims of the earthquake by every means **at their command**.

当地军人动用一切可能的办法帮助地震灾民。

One night, I didn't turn off the water fast enough to her liking. (Para. 1)

P One night, she wasn't pleased with the way I turned off the water, because she thought I didn't do it fast enough.

T 一天晚上，女儿认为我关龙头动作不够快，不乐意了。

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We need to conserve our resources to buy ourselves time so scientists can find new solutions to our problems. (Para. 2)

P We should conserve our resources. If we do so the resources will probably last long enough for scientists to find new solutions to our problems.

T 我们必须节约资源，争取时间，使科学家来得及找到解决问题的新方法。

大学资料都在区网·三号教学楼

His list, in descending order of importance, was: energy, water, food, environment, poverty, terrorism and war, disease, education, democracy, and population. (Para. 4)

P His list contains ten items: energy, water, food, environment, poverty, terrorism and war, disease, education, democracy, and population. They are arranged in such a way that each item is more important than the one that follows it.

T 他列出的问题清单，按重要性排序，依次是：能源、水、食品、环境、贫困、恐怖主义和战争、疾病、教育、民主，以及人口。

The reason energy and water sit at the top, ahead of food and poverty, is that addressing them makes subsequent problems easier to deal with. (Para. 5)

P Energy and water are the top two problems on the list. The reason is that if we start to solve these problems, the other problems that follow them on the list, such as food and poverty, will be easier to deal with.

T 能源和水位居榜首，排在食品和贫困之前，其原因是解决前两个问题，后面的问题就比较容易解决了。

Climate change is expected to intensify droughts and floods while shifting where water will be and when. (Para. 7)

P Climate change will make droughts and floods worse. It will also change where and when rain will fall.

T 气候变化会加剧旱灾和水灾，并改变降水的时间和地点。

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... even though fixing water would help our energy, defense, and health problems. (Para. 9)

P ... even though solving the water problem would help solve energy, defense and health problems.

T ……尽管解决水问题有助于解决能源、国防和健康方面的问题。

大学资料都在公众号：大学资料

Today, according to UNICEF, at least 2.6 billion people lack adequate sanitation, while 1.1 billion people live without clean drinking water period. Water is the key component in all of our daily lives no matter where we live, not only for human use, but for energy, industry, agriculture, and livestock. Now if you look at the Earth, you'll notice that over 75% of our blue planet is water. The problem is only 3% of that water is fresh water.

Fresh water's becoming one of the scarcest resources on the planet. It sounds crazy, right? I mean the world's scientists are saying the oceans are rising and you probably have running water in your house. Shoot. Look at a map and you'll see that 71% of the Earth's surface is covered in water. So how could we be running out?

Well, almost all the water in the world is saltwater, so we can't drink it or grow crops with it. About 1% is in the atmosphere, and only 2.5% is fresh water in liquid form. Out of that small amount of fresh water, most of it is found in icecaps, glaciers, and permanent snow, and almost all the rest is found in the ground. Yep, only about 1% of the world's fresh water is easily available to the creatures and plants and seven billion humans that live on land. That's it.

And here's where we currently stand. Some estimate that we've consumed or are already using about 50% to 60% of our fresh water, 70% going to agriculture, 22% going to industry and about

8% is what we consume in our homes. Except to remember, we eat the food that agriculture grows, and use the things that industry makes, so humans are really doing all of the drinking of the milkshake, so to speak.

And literally, we love our milkshakes. From start to finish, everyday things all around you have a bigger water footprint than you might imagine. For example, it takes about 140 liters of water to produce one cup of coffee, 14,400 liters to produce one kilogram of beef, 24,000 liters to produce one kilogram of chocolate, and 39,090 liters to produce a new car. Bigger said than you thought, right?

About one in eight people don't have access to safe water, and it's even worse in cities where that number is only one in four. On top of that, our population has doubled in the last forty years. Industrialization continues to develop and respond, and our use of water has also quadrupled (变为四倍). Again, to put this simply,

we are literally drinking our home, this lovely green and blue milkshake, all up.

And if you are thinking water scarcity only happens to other people — I've got water. I'm fine. — Think again. Even places that currently have access to clean water are at risk due to things like industrial accidents, agricultural waste, poor water management, natural disasters. And yep, from all of the other things that people like you and I flush down the drain. See that stuff doesn't just go away, it goes back into your milkshake.

Innovations of culture and technology that offer sustainable and cleaner industry, farming and living can change the way that we and all of our descendants live for the better. The way we buy and sell, grow our food, produce our goods and services, and literally govern, develop, build, and interact with everything and everyone around us, matters.

Today, according to UNICEF, at least 2.6 billion people lack adequate sanitation, while 1.1 billion people live without clean drinking water period. Water is the key component in all of our daily lives no matter where we live, not only for human use, but for energy, industry, agriculture, and livestock. Now if you look at the Earth, you'll notice that over 75% of our blue planet is water. The problem is only 3% of that water is fresh water. Over 260 river basins are shared by two or more countries and most of these rivers are without defined legal or institutional arrangements. Consider the Aral Sea, for example, located in central Asia. The name Aral Sea is actually translated into "Sea of Islands", referring to over 1,534 islands that once existed. Today the Aral Sea is down to 10% of its original size. The same thing is happening in other areas that were once flourishing (but) are now turning into desert.

A report titled “Avoiding Water Wars: Water Scarcity and Central Asia’s Growing Importance for Stability in Afghanistan and Pakistan” was recently done by the Senate Foreign Relations Committee. In it, it focuses on rising water tensions that could destabilize central and south Asia. The implications of a water shortage has (have) already caused aggravated demand for agriculture and power generation according to the report. Water, without a doubt, is going to be a huge issue along with oil throughout this century. In the U.S., the world’s largest body of fresh water, the Ogallala Aquifer, an underground lake that extends from the Colorado Rockies to South Dakota going all the way down to the Texas Panhandle with a range of 50 to 300 feet of depth. This water resource is what has made America’s plains the “bread basket” of the world. Unfortunately, like an oil well, this won’t last forever. The Ogallala used to have an average depth of 240 feet. Today its average depth is just 80 feet.

Recently a story done in Texas, about a town named Happy, who has simply run out of water for its farms. What once was a booming town that relied on the Ogallala Aquifer, has now seen its depths of underground water fall between 0–50 feet. Many wells are completely dry and farmers have been forced to hand over their land to the government's Conservation Reserve Program (CRP) in exchange for grants. The U.S. Department of Agriculture recently stated, "The supply is going to run out and the plains will become uneconomical to farm." 60 years is what the U.S. Department of Agriculture gives it. This is a scary thought when you think about the way we have built cities, towns, and homes around something that is simply unsustainable. Consider the Colorado river as well, a river that used to run into the Pacific. Not anymore. By the time the water fills the pools of Vegas and irrigates and provides drinking water throughout the west, not a drop of water makes it to the ocean.

Of course we can't help but comment on how it is the government who has centrally planned people to live in the middle of nowhere, to live in deserts. If a free market would have reigned, we would see a much more practical living condition with very little "middle of the desert living," not only relying on cheap oil, but the relocation of precious water. While researching the water crisis, FutureMoneyTrends.com came across studies about how in some places in Africa so much water has been pulled out of the ecosystem in order to bottle it and send it around the world, some towns have become deserts. As the world population grows, the water crisis will become front page news. According to the consulting company McKinsey & Company, by 2030 global water demand will be 40% greater than today's "accessible, reliable, environmentally sustainable supply." FutureMoneyTrends.com members should have no doubt that this is 100% tied into agriculture and rising food price inflation.

71% of global water withdrawals today go to irrigating our food. The U.S. government has screwed us with subsidies that are handed out to farmers who plant in areas that need an excessive amount of water. Subsidies are also handed out in misguided attempts to turn food into fuel, something that in our opinion is not only driving food costs up, but is wasting precious water resources. Like oil, it takes water for many goods. A cotton T-Shirt, 400 gallons; denim jeans, 1,800 gallons; a car, 39,000 gallons; a board of lumber, 5.4 gallons; a barrel of beer, 1,500 gallons; a gallon of paint, 12 gallons; one ton of steel, 62,000 gallons; a piece of paper, 2.6 gallons. Consider how much water it takes to grow our food, feed our animals, and ourselves.

Water Changes Everything

Water — it's life's most basic need, but there's a water crisis in our world right now, seriously, a crisis. Nearly 1 billion people live without clean drinking water. It's happening all over the world, especially in developing areas of sub-Saharan Africa, Southeast Asia, and Latin America. It's a water crisis because it starts with water, but water affects everything — education, health, poverty, and especially women and children.

Let's look at a family caught in the water crisis. It's likely they live on less than a dollar a day. When they're thirsty, they can't just turn on the faucet for a nice cold glass of water. They don't have a faucet. Instead the women and children go off to collect water. Many walk up to three hours a day to the nearest swamp, pond or river to gather water that's been sitting out in the open, exposed to all kinds of germs.

Time spent gathering water is time they can't spend learning to read, write, earn an income or take care of their family. Some women in sub-Saharan Africa spend more time collecting water than any other activity they do in the day, and the walk isn't just hard; it's dangerous. The women are alone and burdened with 40 pounds of water. Many get hurt; sometimes they're even attacked. When they make it home, the little water they've collected isn't clean. Some families know their water is contaminated with germs that cause diarrhea, dehydration, even death, but what choice do they have?

Kids, especially babies, are affected most by these germs. About every 19 seconds a mother loses one of her children to a water-related illness, and each day almost a billion people are living this way — until they get a little help.

The water crisis is solvable. There are solutions. Some are brand-new and innovative, like water filtration systems; some are age-old, like drilled or hand dug wells. These solutions bring clean water much closer to the people who need it. A safe water project near a village restores hours each day to a person's life. This time — it's opportunity, it's freedom to go to school and get an education, to work or start a business, to raise a family.

Africa alone could save 40 billion hours each year. That's the entire annual workforce of France. Clean water means less disease. That's less money spent on medicine, which means more money for books and school uniforms. And if the water project is built near a school, it can increase attendance, especially among young girls. The water crisis is vast, but we can solve it. Because water changes everything.

Model paper for your reference

There are many things we can do to help conserve clean water supplies. First, we can help keep our water sources clean by persuading people not to throw dirty things into rivers or wells. Besides, since all of us use water every day, we can stop wasting water and learn to do whatever we can to save water in our daily life. And it's not complicated. For example, we should remember to turn off the water as often as we can when doing dishes or taking showers. We should also buy toilets that use less water, and use other water-saving equipment. Perhaps, when we foresee a heavy rain, we can even try to preserve some rainwater and use it to water our plants later. Don't think these efforts are too small. They all help to make our water resources last longer.